TD/TP 1 Supervised learning 2A
2025-2026

Duration 3h
Use of ChatGPT / Claude / other LLMs for part 2 is encouraged.

This session:

- Part 1: Tutorial to work with risk and Bayes classifier
- Part 2: Practical session to review how to generate simulated datasets, load real datasets, and get familiar with the k-nearest neighbors classifier

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Part 1: Bayes rule and risk in binary classification

We focus here on three binary classification problems, detailed below.

Case 1
A sample of size $n = 1000$ is simulated according to the following model:
For $i = 1, \dots, 1000$, $X_i \sim \mathcal{N}(0,1)$, $U_i \sim \mathcal{U}[0,1]$ and $Y_i = \begin{cases} 1(U_i \leq 0.1) & \text{if } X_i \leq 0 \\ 1(U_i > 0.2) & \text{if } X_i > 0 \end{cases}$

Case 2
A sample of size n=1000 is simulated according to the following model:
For $i = 1, \dots, 1000$, $X_i \sim \mathcal{N}(0,1)$, $U_i \sim \mathcal{U}[0,1]$ and $Y_i = \begin{cases} 1(U_i \leq 0.2) & \text{if } X_i \leq 0 \\ 1(U_i > 0.4) & \text{if } X_i > 0 \end{cases}$

Case 3
We work with real data from a survey, based on a random sample of $n = 1000$ coffee consumers.
The target variable Y is binary: "sweetened" or "unsweetened" coffee. The explanatory variable X is the consumer's gender. The dataset has 1000 rows (consumers) and 2 columns (X and Y).
Bivariate statistics show:

- Among women, 20% take sugar in their coffee (Y = "sweetened")
- Among men, 10% take sugar in their coffee (Y = "sweetened")

Questions:

1. For each of the three cases, if possible, provide the Bayes rule and Bayes risk. Detail all steps of the calculation.
2. Is it possible to provide an indicator of the complexity of these problems and rank them accordingly?

Part 2: Data Manipulation and k-Nearest Neighbors Algorithm for Classification

Exercise 1

1. We consider the situation from Case 2 in Part 1:
   1.1. Simulate the described data.
   1.2. Additionally, create another sample of size 200 using the same model. What is the purpose of this new sample?

2. Recall the decision rule associated with the k-nearest neighbors algorithm for classification.

3. Implement the algorithm on the dataset (training, error evaluation) without hyperparameter tuning.

Exercise 2

Now we focus on a real dataset available in the mlbench package.

1. After loading the mlbench package:

   1.1. Load the "Vehicle" dataset, which contains a sample of vehicles. The goal is to predict the type of vehicle (4 categories) based on various shape-related variables.

   1.2. Read the dataset description.

   1.3. How many potential predictors are there in this dataset?

   1.4. Select all predictors except those whose names start with "Skewness" and "Kurtosis".

2. What cost function should be used?

3. Implement the k-nearest neighbors algorithm on the dataset:

   3.1. What is the first mandatory step in any supervised learning method?

   3.2. The k-NN algorithm is particularly sensitive to one aspect. Which one? Based on the analysis in question 1.2, propose a transformation of the dataset.

   3.3. What is the hyperparameter to tune? Propose a strategy to optimize it and implement it numerically. Justify the final chosen value and compute the associated prediction error.